



GA4 Case Study: User Behavior & Conversion Analysis

E-commerce Platform Optimization

PRESENTED BY

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Tool used: Google Analytics

Objectives:



Analyze Traffic Sources

Evaluate acquisition channels to identify high-performing segments and optimize marketing spend.



Understand User Behavior

Map the customer journey to uncover how users interact with key site elements and content.



Identify Conversion Drop-offs

Pinpoint friction in the checkout funnel to reduce abandonment and increase overall conversion rate.



Provide Business Recommendations

Deliver data-driven strategies for platform optimization and long-term business growth.

Data Source & Methodology:



Data Source

Google Analytics Demo Account representing an e-commerce platform.



Tools & Method

Analyzed using Google Analytics and visualized via Excel to derive actionable insights.

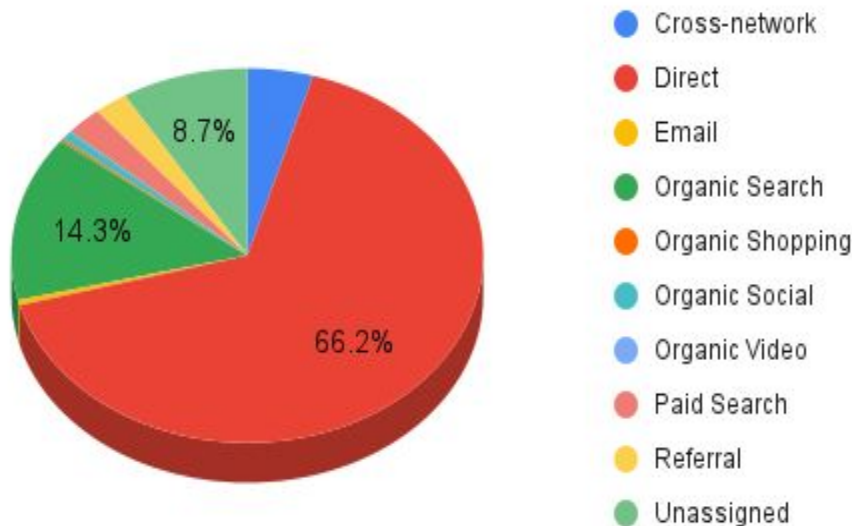


Metrics Used

- **Sessions:** Measure of traffic volume.
 - **Engagement Rate:** Indicator of user interaction quality.
 - **Item Purchase:** Core conversion indicator.
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Traffic Distribution Analysis:

% Users by Channel



Key Acquisition Insights

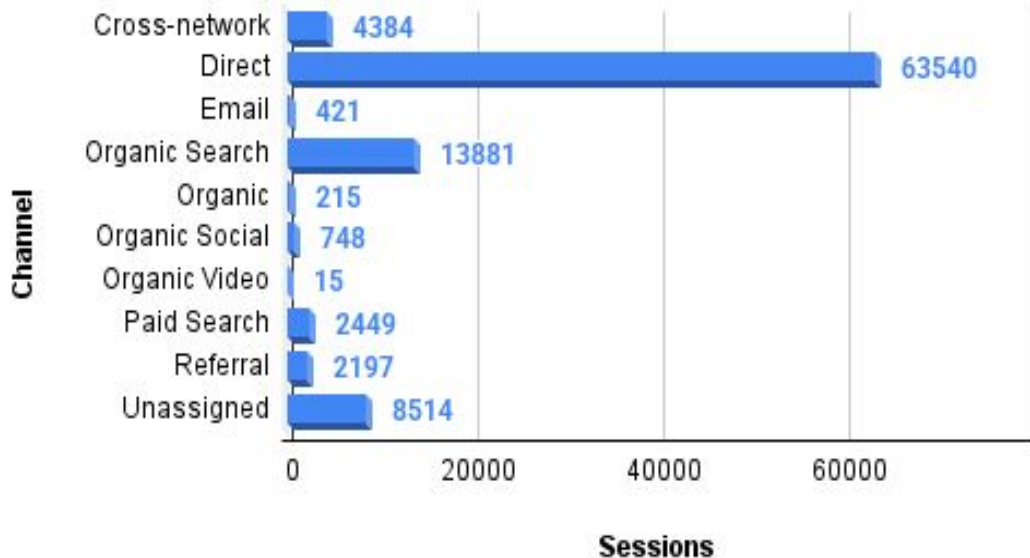
Organic Search (~14.3%) and **Direct (~66.2%)** channels dominate the traffic landscape, accounting for over 80% of total users.

This distribution indicates **strong search visibility** and **high brand recall** among the target audience.

Strategic Note: While volume is high, traffic alone does not guarantee conversions. It is critical to evaluate channel efficiency to optimize ROI.

Sessions by Channel Analysis:

Acquisition Volume per Channel



Strategic Observations

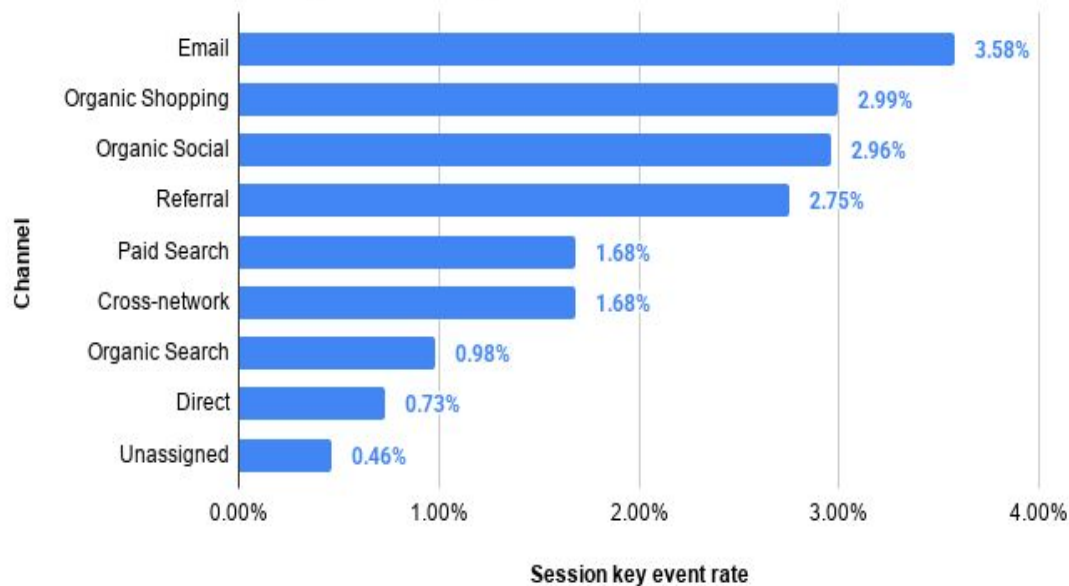
Brand Loyalty: "Direct" traffic (63.5k sessions) shows strong recall; focus on retention and CRM.

Tracking Error: High "Unassigned" volume (8.5k sessions) suggests a configuration issue; prioritize a UTM tagging audit to fix attribution.

Diversification: Small Email and Social traffic indicates growth opportunity to capture new audiences.

Conversion Rate Analysis:

Conversion Rate by Marketing Channel



Conversion Insights

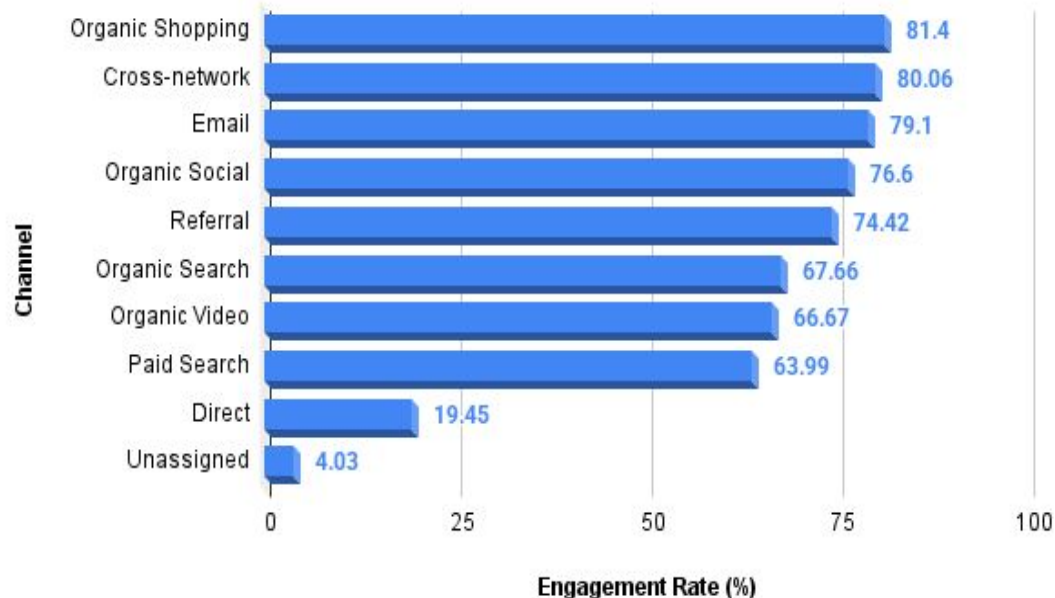
Email Performance: Email is the top performer, converting at 3.58% and driving high-intent purchases.

Direct Traffic Gap: Direct traffic drives high volume but has a low conversion rate (0.73%), requiring landing page optimization.

ROI Strategy: Shift resources to high-converting channels like Email and Organic Shopping to maximize ROI.

Engagement Rate by Channel Analysis:

Engagement Rate(%) vs. Channel



Engagement Insights

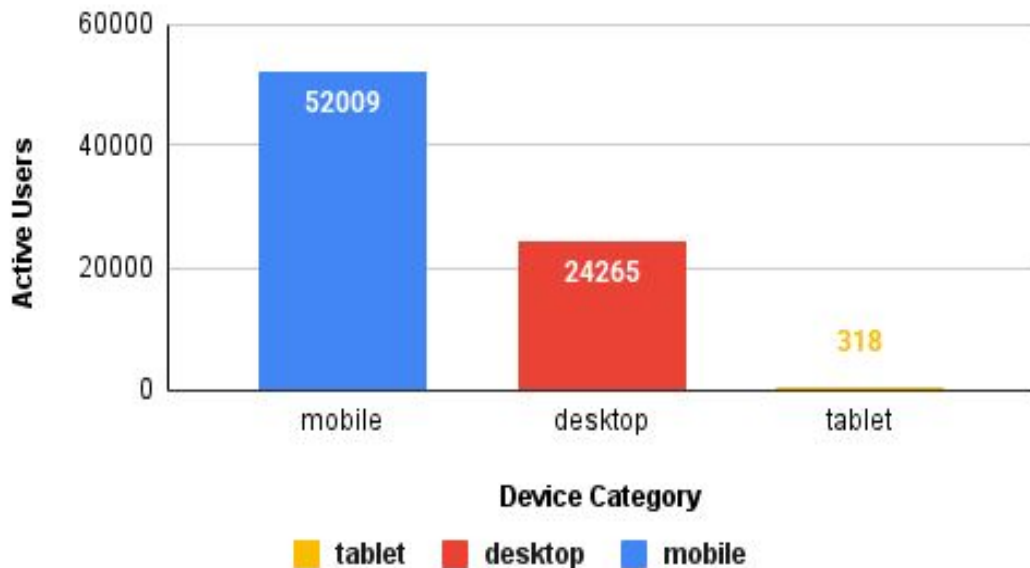
High Fit Channels: "Organic Shopping" (81.4%) and "Email" (79.1%) show strong content-market fit, proving high relevance for these audiences.

"Direct" Challenge: Despite the highest volume, "Direct" traffic engagement is very low (19.45%), highlighting a critical need for landing page optimization.

Optimization Strategy: Replicate successful content structure from top-performing channels to improve low-engagement segments.

Device Analysis:

Active Users by Device Category



User Base Insights

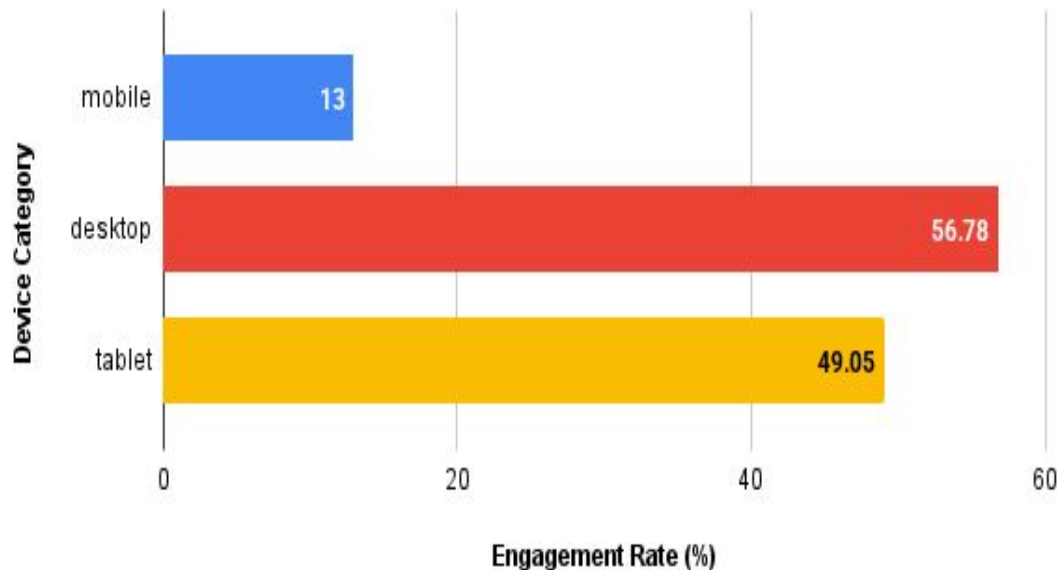
Mobile-First Dominance: Mobile users (52,009) dominate the active user base, requiring a mobile-first UI/UX strategy.

Platform Prioritization: Despite solid desktop usage (24,265), new features must be optimized for mobile performance first.

Resource Efficiency: Deprioritize negligible tablet usage (318) to reallocate engineering resources to mobile and desktop improvements.

Engagement Rate by Device:

Engagement Rate (%) by Device Category



Critical Engagement Gaps

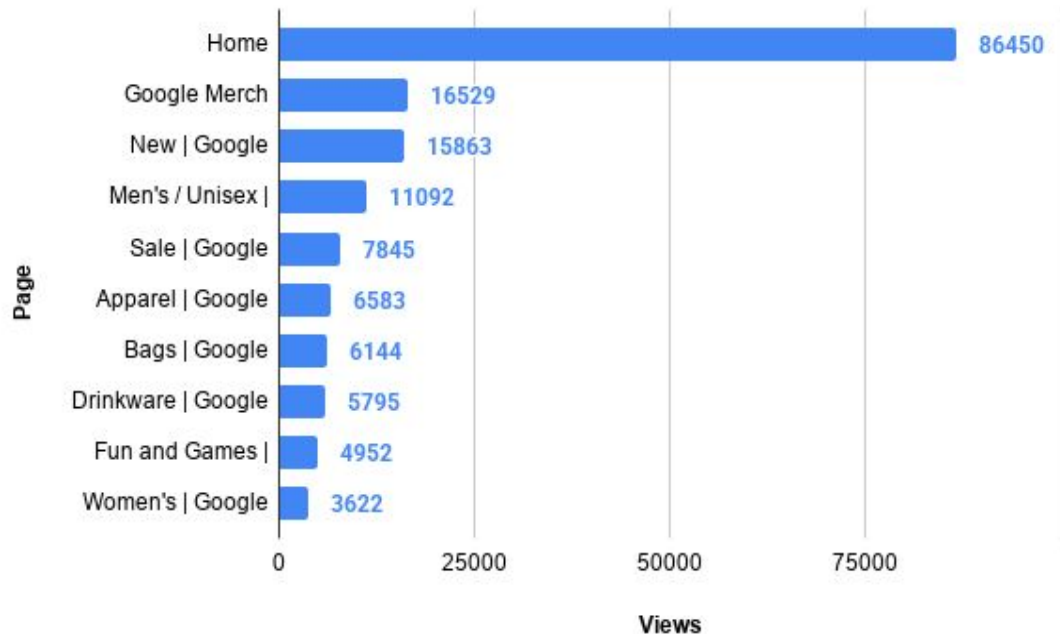
Critical Engagement Gap: There is a severe performance disparity; mobile engagement (13%) is drastically lower than desktop (56.78%).

UX Red Flag: Users are visiting on mobile but failing to interact, suggesting significant friction (e.g., slow load times or poor navigation).

Immediate Priority: Clear business case for a "mobile-first" UX audit to align mobile performance with desktop standards.

Top Pages by Views Analysis:

Top 10 Pages by Views



Analysis Insights

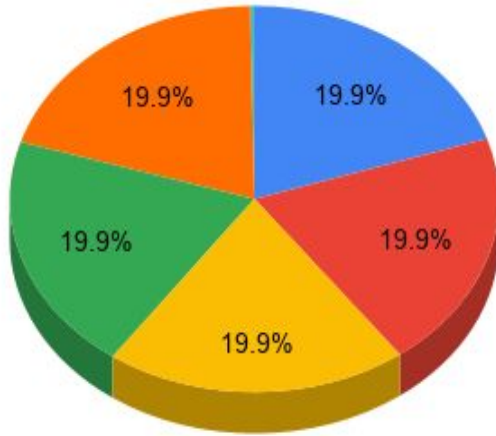
Centralized Traffic: The "Home" page acts as the primary gateway (86.4k views); it is the single most critical asset for user acquisition.

Core Product Interest: High volume on "Google Merch" and "New" pages confirms strong user intent for core brand offerings and latest releases.

Navigation Opportunity: Capitalize on the Home page's massive reach by optimizing internal links to better funnel traffic into specific product categories and drive conversions.

Engagement Rate by Page Analysis:

Engagement Rate % Distribution



● Android ● Gemini ● Google Cloud ● Google Merch Shop ● Home ● (not set)

Analysis Insights

Balanced Content Performance: Chart shows even distribution of user engagement across core content pillars.

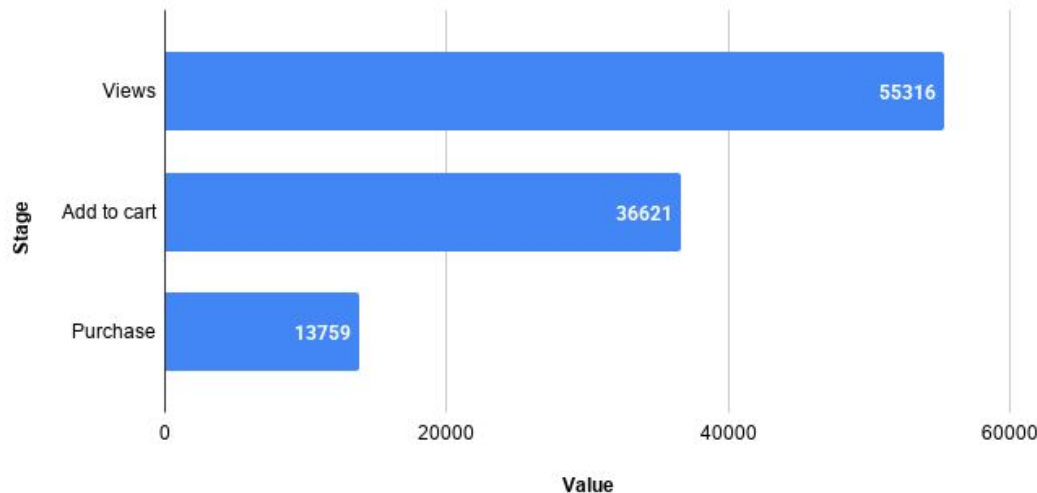
Content Portfolio Analysis: Mapping engagement helps determine if traffic is concentrated on "hero" pages or evenly distributed.

Strategic Benchmarking: This baseline view will be used to identify high-performing pages and replicate successful layouts.

Funnel Analysis:

User Conversion Journey

Value by Stage Funnel



Funnel Insights

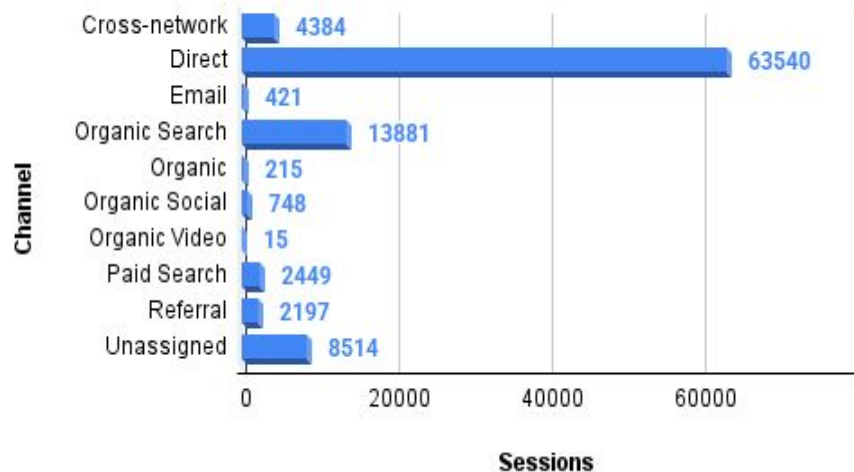
The Cart-to-Purchase Leak: Critical drop-off between "Add to cart" (36,621) and "Purchase" (13,759), losing ~63% of high-intent users.

Optimization Priority: Representing the largest revenue opportunity. Streamlining checkout with fewer forms and guest options is vital.

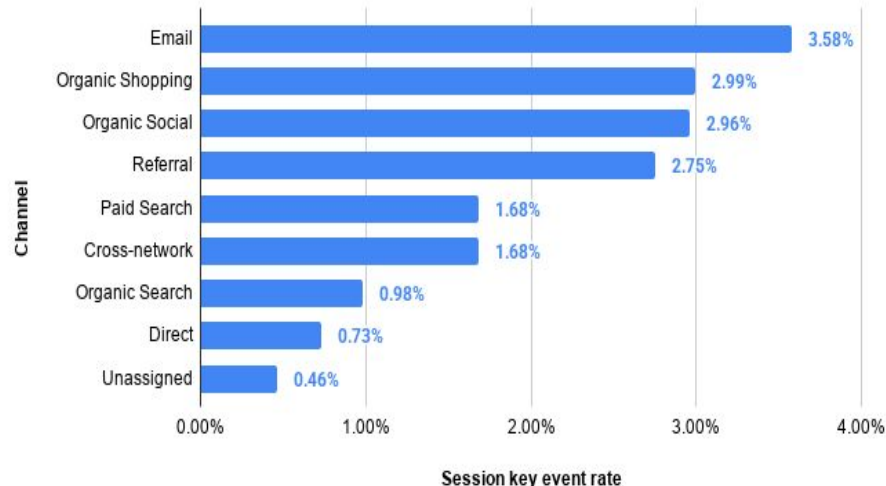
Strategic Goal: Current conversion is ~25%. Future A/B testing must target the cart-to-checkout flow to drive growth.

Conversion Rate Summary:

Sessions by Channel



Conversion Rate by Marketing Channel



Channels with high traffic volume do not necessarily yield high conversion efficiency, highlighting the importance of targeting high-intent users.

"High Traffic, Low Conversion": Organic | **"Low Traffic, High Conversion":** Email

Key Findings & Analysis:



Traffic Composition

Direct and Organic channels drive 60%+ of total traffic; however, "Unassigned" traffic (8.5k sessions) signals a critical attribution error.



The Engagement Gap

Mobile accounts for ~70% of users, but engagement (13%) drastically underperforms compared to Desktop (56.78%).



Funnel Leakage

High drop-off rate (~75%) between "Add to Cart" and "Purchase" stages indicates significant friction in the checkout process.

Strategic Insights:



Volume vs. Value

We have strong brand awareness (high Direct traffic), but we are failing to monetize this effectively due to UX friction.



Conversion Inefficiency

While Direct/Organic brings volume, Email and Organic Shopping are our most efficient conversion engines (~3.5%).



Data Reliability

Tracking gaps like "Unassigned" and "(not set)" hinder our ability to make data-driven decisions on marketing ROI.

Recommendations:



Mobile-First UX Audit

Immediate overhaul of mobile navigation and load times to close the engagement gap (Target: 10–15% uplift).



Checkout Optimization

Streamline the payment flow and cart-to-purchase path to reduce friction and improve conversion from ~37% to 50%+.



Data Integrity & Scaling

Execute a full UTM tagging audit to fix attribution; concurrently reallocate spend toward Email and Shopping channels.

Conclusion:



Shift in Strategy

Transition from a volume-centric traffic model to an efficiency-focused conversion model.



Foundational Focus

Prioritizing Data Hygiene and Mobile UX will serve as the engine for sustainable, measurable growth.



Final Impact

Stabilize the funnel and create a scalable, data-verified platform for long-term revenue.